

Problem 20

Evaluate the integral:

$$\int (m \sinh x + n \cosh x) dx,$$

where m and n are constants.

Solution

We are tasked with evaluating the integral:

$$\int (m \sinh x + n \cosh x) dx.$$

Step 1: Use the standard integrals of $\sinh x$ and $\cosh x$

Recall the following standard integrals:

$$\int \sinh x dx = \cosh x \quad \text{and} \quad \int \cosh x dx = \sinh x.$$

Using these, we can break the integral into two parts.

Step 2: Split the integral

We can now split the given integral as follows:

$$\int (m \sinh x + n \cosh x) dx = m \int \sinh x dx + n \int \cosh x dx.$$

Step 3: Solve each integral

Now, we integrate each term:

$$m \int \sinh x dx = m \cosh x$$

and

$$n \int \cosh x dx = n \sinh x.$$

Step 4: Combine the results

Combining the results from the two integrals, we get:

$$\int (m \sinh x + n \cosh x) dx = m \cosh x + n \sinh x + C,$$

where C is the constant of integration.

Final Answer

The solution to the integral is:

$$\int (m \sinh x + n \cosh x) dx = m \cosh x + n \sinh x + C.$$